DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING (AI & ML, IOT)

VISION OF THE DEPARTMENT

To emerge as an epicentre having transformative impact in the education and research of Artificial Intelligence & Machine Learning and Internet of Things with an ecosystem that contributes to the society through innovativeness and creativeness.

MISSION OF THE DEPARTMENT

- To Impart innovative teaching and learning methodologies to generate knowledge through the state-of-the-art concepts and technologies in AI-ML and IoT.
- To produce successful Computer Science and Engineering graduates with a specialization in AI-ML and IoT with personal and professional responsibilities and commitment to lifelong learning with societal aspirations.
- To establish centres of excellence in leading areas of AI-ML and IoT in collaboration with industry to uplift innovative research and development.

B. TECH.

CSE (INTERNET OF THINGS)

B. TECH. CSE (IoT)

PROGRAM EDUCATIONAL OBJECTIVES

PEO-I: Apply core principles of Computer Science with specialization in IoT to design and develop sustainable solutions addressing complex real-world challenges.

PEO-II: Demonstrate teamwork and leadership with ethical values and adaptability while engaging in lifelong learning of emerging IoT technologies.

PEO-III: Foster innovation, research, and entrepreneurship by applying creativity and problem-solving skills to support industry, start-ups, and national development.

B. TECH. CSE (IoT)

PROGRAM OUTCOMES

PO-1: **Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO-2: **Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).

PO-3: **Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).

PO-4: **Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO-5: **Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).

PO-6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO-7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).

PO-8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO-9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO-10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO-11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

B. TECH. CSE (IoT)

PROGRAM SPECIFIC OUTCOMES

PSO-1: Apply Computer Science fundamentals along with IoT-specific and emerging technologies to design effective IoT solutions.

PSO-2: Design and integrate secure IoT-based hardware–software systems with industry readiness and project management skills.

PSO-3: Develop and deploy sustainable IoT applications through research, innovation, and collaboration with industry and academia.

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD

B. TECH. I YEAR

CSE (INTERNET OF THINGS)

I SEMESTER **R25**

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT101	Matrices and Calculus	3	1	0	4	4
25BS1PH102	Advanced Engineering Physics	3	0	0	3	3
25HS1EN101	English for Skill Enhancement	3	0	0	3	3
25ES1CS101	Programming for Problem Solving	3	0	0	3	3
25ES1IN101	Introduction to Internet of Things	3	0	0	3	3
25BS2PH102	Advanced Engineering Physics Laboratory	0	0	2	2	1
25ES2CS101	Programming for Problem Solving Laboratory	0	0	2	2	1
25HS2EN101	English Language and Communication Skills Laboratory	0	0	2	2	1
25ES2IT101	IT Workshop	0	0	2	2	1
25MN6HS101	Induction Programme	2	0	0	2	0
Total		**17**	**1**	**8**	**26**	**20**

II SEMESTER **R25**

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT106	Ordinary Differential Equations, Laplace Transforms and Vector Calculus	2	1	0	3	3
25BS1CH101	Chemistry for Engineers	3	0	0	3	3
25ES1EE102	Basic Electrical and Electronics Engineering	3	0	0	3	3
25ES1IT102	Data Structures	3	0	0	3	3
25ES3ME101	Engineering Drawing	0	0	6	6	3
25BS2CH101	Engineering Chemistry Laboratory	0	0	2	2	1
25ES2IT102	Data Structures Laboratory	0	0	2	2	1
25ES2CY101	Python Programming Laboratory	0	0	2	2	1
25ES2ME101	Engineering Workshop	0	0	2	2	1
25ES2EE102	Basic Electrical and Electronics Engineering Laboratory	0	0	2	2	1
25MN6HS102	Environmental Science	2	0	0	2	0
Total		**13**	**1**	**16**	**30**	**20**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS1MT101) MATRICES AND CALCULUS

TEACHING SCHEME		
L	T/P	C
3	1	4

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematical Knowledge at Pre-University level

COURSE OBJECTIVES:
- To learn the concept of a rank of the matrix and applying this concept to check the consistency of a system of linear equations
- To learn the concept of eigen values and eigen vectors and to reduce the quadratic form to canonical form
- To know the geometrical approach to the mean value theorems and their application to the mathematical problems
- To find the maxima and minima of function of two and three variables
- To evaluate multiple integrals and their applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Compute the rank of a matrix and analyze the solution of a system of linear equations
CO-2: Calculate the Eigenvalues and Eigen vectors and reduce the quadratic form to canonical form using orthogonal transformations
CO-3: Solve the applications on the mean value theorems
CO-4: Find the extreme values of functions of two and three variables with/ without constraints
CO-5: Evaluate the multiple integrals and apply the concept to find areas, volumes

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	1	1	1	-	2	1	2	2	1	-
CO-2	3	2	2	1	2	1	1	-	2	1	2	2	1	-
CO-3	3	2	2	1	1	1	1	-	2	1	2	2	1	-
CO-4	3	2	2	1	2	1	1	-	2	1	2	2	1	-
CO-5	3	2	2	1	2	1	1	-	2	1	2	2	1	-

UNIT-I:
Matrices: Rank of a matrix by Echelon form and Normal form – Inverse of Non-singular matrices by Gauss-Jordan method. System of linear equations: Solving system of Homogeneous and Non-Homogeneous equations, Gauss Seidel Iteration Methods.

UNIT-II:
Eigenvalues and Eigenvectors: Eigenvalues – Eigenvectors and their properties – Diagonalization of a matrix – Cayley-Hamilton Theorem (without proof) – finding inverse and power of a matrix by Cayley-Hamilton Theorem – Quadratic forms and

Nature of the Quadratic Forms – Reduction of Quadratic form to canonical forms by Orthogonal Transformation.

UNIT-III:

Single Variable Calculus: Limits and Continuous function and its properties. Mean value theorems: Rolle's theorem – Lagrange's Mean value theorem with their Geometrical Interpretation and applications – Cauchy's Mean value Theorem – Taylor's Series (All the theorems without proof).

Curve Tracing: Curve tracing in cartesian and polar coordinates

UNIT-IV:

Partial Differentiation and Its Applications: Definitions of Limit and continuity – Partial Differentiation: Euler's Theorem – Total derivative – Jacobian – Functional dependence & independence. Applications: Maxima and minima of functions of two variables and three variables using method of Lagrange multipliers.

UNIT-V:

Multiple Integrals: Evaluation of Double Integrals (Cartesian and polar coordinates) – change of order of integration (only Cartesian form) – Change of variables for double integrals (Cartesian to polar). Evaluation of Triple Integrals – Change of variables for triple integrals (Cartesian to Spherical and Cylindrical polar coordinates). Applications: Areas by double integrals and volumes by triple integrals.

TEXT BOOKS:

1. Higher Engineering Mathematics, B. S. Grewal, 36th Edition, Khanna Publishers, 2010
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, 5th Editon, Narosa Publications, 2016

REFERENCES:

1. Advanced Engineering Mathematics, Erwin Kreyszig, 13th Edition, John Wiley & Sons, 2006
2. Thomas' Calculus, G. B. Thomas, 13th Edition, Pearson, 2014
3. A Text book of Engineering Mathematics, N. P. Bali and Manish Goyal, Laxmi Publications, Reprint, 2008
4. Higher Engineering Mathematics, H. K. Dass and Er. Rajnish Verma, S. Chand

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS1PH102) ADVANCED ENGINEERING PHYSICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: 10 + 2 Physics

COURSE OBJECTIVES:
- To explore the working and applications of lasers and fibre optics in modern technology
- To understand fundamental concepts of quantum mechanics and their applications in solids
- To explain various types of semiconductors, semiconductor devices and nanomaterials
- To learn the properties and applications of dielectric and magnetic materials
- To introduce quantum computing principles and quantum gates

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Explain the principles of lasers and fibre optics and their applications in communication and sensing
CO-2: Apply quantum mechanical principles to explain particle behaviour and energy band formation in solids
CO-3: Identify the role of semiconductor devices and nanomaterials in science and engineering applications
CO-4: Classify magnetic and dielectric materials and explain their properties, synthesis, and applications
CO-5: Understand quantum computing concepts and quantum gates

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	-	-	2	1	-	-	-	1	2	-	-
CO-2	3	3	-	-	-	-	1	-	-	-	1	1	-	-
CO-3	3	2	-	-	-	2	1	-	-	1	1	-	-	-
CO-4	3	3	1	-	-	2	1	-	-	1	1	1	-	-
CO-5	3	3	1	2	1	-	1	-	-	1	1	-	-	-

UNIT-I:
Laser and Fibre Optics: Introduction to laser, characteristics of laser, Einstein coefficients and their relations, metastable state, population inversion, pumping, lasing action, Ruby laser, He-Ne laser, Semiconductor laser, applications: Bar code scanner, LIDAR for autonomous vehicle.
Introduction to fibre optics, total internal reflection, construction of optical fibre, acceptance angle, numerical aperture, classification of optical fibres, losses in optical

fibre, applications: optical fibre for communication system, sensor for structural health monitoring.

UNIT-II:

Quantum Mechanics and Band Theory of Solids: Introduction, de-Broglie hypothesis, Heisenberg uncertainty principle, physical significance of wave function, postulates of quantum mechanics: operators in quantum mechanics, eigen values and eigen functions, expectation value; Schrödinger's time independent wave equation, particle in a 1D box.

Introduction to Band theory, Bloch's theorem (qualitative), Kronig-Penney model (qualitative): E-k diagram, effective mass of electron, formation of energy bands, classification of solids.

UNIT-III:

Semiconductors & Nanotechnology: Intrinsic and Extrinsic Semiconductors (Qualitative), Hall Effect-Hall coefficient and its applications, Direct and Indirect band gap semiconductors, Construction, Working principle, V-I characteristics and applications of LED and Solar Cell.

Introduction to nanoscience and nanotechnology, surface to volume ratio, quantum confinement, synthesis methods: Ball milling, sol-gel method, Characterization Techniques: X-ray diffraction (XRD)- calculation of average crystallite size using Debye Scherrer's formula and Scanning Electron Microscopy (SEM), applications of nanomaterials.

UNIT-IV:

Dielectric and Magnetic Materials: Introduction to dielectric materials, types of polarization (qualitative): electronics, ionic & orientation; ferroelectric, piezoelectric, pyroelectric materials and their applications: Ferroelectric Random-Access Memory (Fe-RAM), load cell and fire sensor.

Introduction to magnetic materials, classification of magnetic materials, hysteresis, Weiss domain theory of ferromagnetism, soft and hard magnetic materials, applications: magnetic hyperthermia for cancer treatment, magnets for EV, Giant Magneto Resistance (GMR) device.

UNIT-V:

Quantum Computing: Introduction, concept of quantum computer, classical bits, Qubits, multiple Qubit system, quantum computing system for information processing, Dirac's Bra and Ket notation and their properties.

Hilbert space, Bloch's sphere, introduction to entanglement and Quantum gates, challenges and advantages of quantum computing over classical computation.

TEXT BOOKS:

1. Textbook of Engineering Physics, M. N. Avadhanulu, P. G. Kshirsagar & T. V. S. Arun Murthy, 11th Edition, S. Chand Publications, 2019
2. Introduction to Solid State Physics, Charles Kittel, John Wiley & Sons
3. Introduction to Classical and Quantum Computing, Thomas G. Wong, Rooted Grove

REFERENCES:

1. Quantum Computation and Quantum Information, Michael A. Nielsen & Isaac L. Chuang, Cambridge University Press
2. Optical Fiber Communications Principles and Practice, John M. Senior, Pearson Education Limited
3. Engineering Physics, B. K. Pandey and S. Chaturvedi, 2^{nd} Edition, Cengage Learning, 2022
4. Engineering Physics, P. K. Palanisamy, SciTech Publications
5. Concepts of Modern Physics, Aurther Beiser, McGraw-Hill

ONLINE RESOURCES:

1. https://shijuinpallotti.wordpress.com/wp-content/uploads/2019/07/optical-fiber-communications-principles-and-pr.pdf
2. https://www.geokniga.org/bookfiles/geokniga-crystallography_0.pdf
3. https://dpbck.ac.in/wp-content/uploads/2022/10/Introduction-to-Solid-State-PhysicsCharles-Kittel.pdf
4. https://www.thomaswong.net/introduction-to-classical-and-quantum-computing-1e4p.pdf

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25HS1EN101) ENGLISH FOR SKILL ENHANCEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Basic English

COURSE OBJECTIVES:
- To enhance vocabulary through word formation processes
- To read and comprehend different kinds of texts
- To write clear, concise paragraphs, essays, and letters to produce appropriate technical prose
- To improve coherence and cohesion in writing and speaking
- To recognize and practice the use of rhetorical elements necessary for the successful practice of scientific and technical communication

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Use vocabulary contextually and effectively in oral and written communication
CO-2: Demonstrate an understanding of the rules of functional Remedial Grammar and sentence structures
CO-3: Demonstrate improved competence in Standard Written English, including Remedial Grammar, sentence and paragraph structure, and coherence, and use this knowledge to accurately communicate technical information
CO-4: Apply principles of writing while developing paragraphs, essays, letters and technical prose
CO-5: Employ appropriate rhetorical patterns of discourse in scientific and technical communication

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	1	1	1	1	-	1	1	1	2	3	2	-	-	-
CO-2	2	2	2	2	2	2	2	1	3	3	2	-	-	-
CO-3	2	2	2	2	1	3	2	1	3	3	2	-	-	-
CO-4	1	1	1	1	1	2	2	1	2	3	2	-	-	-
CO-5	1	1	1	1	-	2	1	1	2	2	1	-	-	-

UNIT-I:
1. **Reading:** -The Generation Gap' by Benjamin M. Spock
   The Lotos Eaters – Alfred Tennyson, The Land of High Passes – Bhagyalakshmi Krishnamurthy
2. **Grammar:** Degrees of comparisons, Conjunctions, and Prepositions – Identifying Common Errors
3. **Vocabulary:** Word Formation (Affixation, Compounding, Conversion, Blending, Borrowing)

4. **Writing:** Punctuation, Clauses and Sentences, Transitional Devices, Paragraph Writing- Process

UNIT-II:
1. **Reading:** Emerging Technologies
   Medicate – Oleksandr Gorpynich Of Studies – Francis Bacon
2. **Grammar:** Articles Noun-Pronoun Agreement; Concord
3. **Vocabulary:** Word Formation (Prefixes, Suffixes, Root Words)
4. **Writing:** Essay Writing -Descriptive, Argumentative, Expository

UNIT-III:
1. **Reading: Leisure- William Henry Davies -Be Thankful - Unknown Author**
   The Rising of the Moon – Lady Gregory, With The Photographer – Stephen Leacock
2. **Grammar:** Misplaced Modifiers
3. **Vocabulary:** Synonyms and Antonyms
4. **Writing:** Letter Writing- Formal Letter Writing– Letter of Complaint, Letter of Requisition, Email Writing, Email Etiquette

UNIT-IV:
1. **Reading:** -Why a Start-Up Needs to Find its Customers First'- Pranav Jain
   Too dear – Leo Tolstoy, Shooting an Elephant – George Orwell
2. **Grammar:** Clichés, Redundancies
3. **Vocabulary:** Common Abbreviations
4. **Writing:** Summary Writing, Job Application, Resume Writing

UNIT-V:
1. Reading: **'Professional Ethics'**
2. Patterns of Writing: Comparison and Contrast
3. Patterns of Writing: Cause and Effect Paragraphs
4. Patterns of Writing: Classification Paragraph
5. Patterns of Writing: Problem–Solution Pattern of Writing

TEXT BOOKS:
1. Board of Editors, English for the Young in the Digital World, Orient Black Swan, 2025
2. Unlocking English: A Skill-Based Approach for Language Proficiency
3. Technical Communication, Rebecca E. Burnett, 6th Edition, Cengage Learning

REFERENCES:

1. Practical English Usage, Swan Michael, New Edition, Oxford University Press, 2016
2. English Grammar Just for You, Karal, Rajeevan, Oxford University Press, 2023
3. Empowering with Language: Communicative English for Undergraduates. Cengage Learning India, 2024
4. Communication Skills – A Workbook, Sanjay Kumar & Pushp Lata, Oxford University Press, 2022
5. Study Writing, Hamp, Liz., Lyons and Heasly, Ben, C U P, 2006

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES1CS101) PROGRAMMING FOR PROBLEM SOLVING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To relate basics of programming language constructs and problem-solving techniques
- To classify and implement control structures and derived data types
- To analyze and develop effective modular programming
- To construct mathematical problems and real time applications using C Language

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Design solutions for the given problem statements
CO-2: Develop programs using language specific constructs for solving computational problems efficiently
CO-3: Apply the modular approach for solving real time problems
CO-4: Process diverse data using derived types along with user-defined types
CO-5: Demonstrate input/output operations and error management for persistent data storage and retrieval

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	-	3	3	-	1	-	-	-	-	-	-	3	1	1
CO-2	3	2	3	1	2	-	-	-	1	-	1	3	2	1
CO-3	3	2	3	1	2	-	-	1	1	-	1	2	2	1
CO-4	3	2	3	1	2	-	-	1	1	-	1	2	1	2
CO-5	3	2	3	1	3	1	1	-	1	-	1	1	2	2

UNIT-I:

Introduction to Programming: Compilers, compiling and executing a program. Representation of Algorithm, Flowchart/ Pseudocode with examples, Program design and structure of C programming. Variables, Data types, Operators and expressions, precedence and associativity, Expression evaluation, type conversion.
I/O: Simple input and output with scanf() and printf(), formatted I/O.
Conditional Branching: Branching with if, if-else, nested if-else, else-if ladder, switch case.

UNIT-II:

Loops, Arrays, Strings:
Loops: Iteration with while, do-while, for, and Nested loops. Jump statements: break, continue, goto, and return.
Arrays: Single dimensional and Multidimensional arrays, Declaring, Initializing, accessing and manipulating elements of arrays.

Strings: Introduction to strings, handling strings as array of characters, string built-in functions, array of strings.

UNIT-III:

Searching, Sorting, Functions:

Searching: Linear search and Binary search techniques.

Sorting: Basic algorithms to sort array of elements (Bubble, Insertion and Selection sort algorithms), Basic concept of order of complexity through the example programs

Functions: Designing structured programs, declaring a function, Signature of a function, Parameters and return type of a function, passing parameters to functions. Recursion with examples. C library functions, Storage classes.

UNIT-IV:

Structures and Pointers:

Structures: Defining structures, initializing structures, unions, Array of structures, Bit Fields.

Pointers: Defining pointers, Pointers to Arrays, Strings and Structures, Passing arrays and structures to functions. Dynamic memory allocation, self-referential structures.

UNIT-V:

Preprocessor Directives and File Handling in C:

Preprocessor Directives: Symbolic constants, macro expansion and file inclusion. User Defined Data Types: enum, typedef.

Files: Text and Binary files, file input/output operations, stdin, stdout and stderr. Error Handling in Files, random access of files, command line arguments.

TEXT BOOKS:

1. The C Programming Language, Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall of India
2. C Programming and Data Structures, B. A. Forouzan and R. F. Gilberg, 3rd Edition, Cengage Learning
3. C: The Complete Reference, Herbert Schildt, 4th Edition, McGraw-Hill

REFERENCES:

1. Problem Solving and Program Design in C, Jeri R. Hanly and Elliot B. Koffman, 7th Edition, Pearson
2. Computer Fundamentals and C, E. Balagurusamy, 2nd Edition, McGraw-Hill
3. Let us C, Yashavant Kanetkar, 18th Edition, BPB
4. How to Solve it by Computer, R. G. Dromey, 16th Impression, Pearson
5. Programming in C, Stephen G. Kochan, 4th Edition, Pearson Education

ONLINE RESOURCES:

1. https://nptel.ac.in/courses/106105171
2. https://ugcmoocs.inflibnet.ac.in/index.php/courses/view_ug/307

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES1IN101) INTRODUCTION TO INTERNET OF THINGS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To understand the basics of Internet of Things
- To impart knowledge of components of Internet of Things
- To explore the methodologies for IoT Systems
- To provide foundational knowledge and hands-on experience in using Arduino hardware and software for basic sensor and actuator interfacing
- To introduce security requirements, challenges, and solutions in managing an IoT ecosystem

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Establish knowledge in a concise manner how the Internet of things work
CO-2: Illustrate various enabling technologies for IoT
CO-3: Identify and interpret design methodology of IoT platform
CO-4: Implement simple input/output operations and establish serial communication using Arduino
C0 -5: Analyze security threats in IoT components and propose suitable security measures

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	1	2	-	-	-	-	1	-	2	3	2
CO-2	3	2	2	-	2	1	1	-	-	1	-	2	3	2
CO-3	2	3	3	2	2	-	1	-	-	1	1	2	3	3
CO-4	2	2	3	2	3	-	-	-	1	2	1	2	3	3
CO-5	2	3	2	2	2	2	2	3	-	2	1	2	2	3

UNIT-I:
Introduction to Internet of Things (IoT): IoT definition, Characteristics of IoT, Physical Design of IoT: Things in IoT, IoT Protocols, Logical Design of IoT: IoT Functional Blocks, IoT Communication Models, IoT Communication APIs.

UNIT-II:
IoT Enabling Technologies: Wireless Sensor Networks, Cloud Computing, Big Data Analytics, Communication Protocols, Embedded Systems. IoT Levels-Level 1 to Level 6.

UNIT-III:
IoT Design Methodology: Purpose & Requirements Specification, Process Specification, Domain Model Specification, Information Model Specification, Service Specifications, IoT Level Specification, Functional View Specification, Operational

View Specification, Device & Component Integration, Application Development. Case Study on IoT system for Weather monitoring.

UNIT-IV:

Really Getting Started with Arduino: The Arduino Hardware, The Software (IDE), Steps to Install Arduino IDE, Port Identification: Windows, Sensors and Actuators, Blinking an LED, Using a Pushbutton to Control the LED, Trying Out Other On/Off Sensors, Controlling Light with PWM, Analogue Input, Serial Communication.

UNIT-V:

Security Management of an IoT Ecosystem: Introduction, Security Requirements of an IoT Infrastructure, Authentication, Authorization, and Audit Trial (AAA) Framework, Security Concerns of Cloud Platforms, Security Threats of Big Data, Security Threats in Smartphones, Security Solutions for Mobile Devices, Security Concerns in IoT Components, Security Measures for IoT Platforms/Devices.

TEXT BOOKS:

1. Internet of Things, A Hands on Approach, Vijay Madisetti, Arshdeep Bahga, University Press
2. The Internet of Things: Enabling Technologies, Platforms, and Use Cases, Pethuru Raj and Anupama C. Raman, CRC Press
3. Internet of Things with Raspberry Pi and Arduino, Boca Raton, Singh R., Gehlot A., Gupta L., Singh B., Swain M., CRC Press, 2020

REFERENCES:

1. Internet of Things for Architects: Architecting IoT Solutions by Implementing Sensors, Communication Infrastructure, Edge Computing, Analytics, and Security, Perry Lea, Packt Publishing, 2018
2. Internet of Things Security Principles, Applications, Attacks, and Countermeasures B. B. Gupta Megha Quamara, CRC Press, 2018
3. Getting Started with Arduino, Massimo Banzi, 1st Edition, O'Reilly, 2009

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS2PH102) ADVANCED ENGINEERING PHYSICS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To explore the working principles of lasers and optical fibers through hands-on experiments
- To analyze the characteristics of semiconductors and devices
- To study resonance and magnetic induction in electromagnetic systems
- To synthesize the magnetic materials and study magnetic, dielectric properties of materials
- To develop skills in data analysis, interpretation, and scientific reporting

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Verify laser and optical fibre parameters through experimental study
CO-2: Characterize semiconductors and devices such as LED, laser diode etc
CO-3: Demonstrate resonance phenomena and realize tangent law of magnetism
CO-4: Synthesize magnetite nanomaterial, analyse magnetic and dielectric properties of materials
CO-5: Apply scientific methods for accurate data collection, analysis, and technical report writing

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-2	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-3	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-4	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-5	3	3	1	2	2	2	1	3	3	2	1	-	2	-

LIST OF EXPERIMENTS:

(Any 8 experiments are to be performed)

1. a) Determination of wavelength of a laser using diffraction grating.
   b) Study of V-I & L-I characteristics of a given laser diode.
2. a) Determination of numerical aperture of a given optical fibre
   b) Determination of bending losses of a given optical fibre.
3. Synthesis of magnetite (Fe3O4) powder using sol-gel method
4. Determination of energy gap of a semiconductor.
5. Magnetic field along the axis of current carrying coil – Stewart and Gee
6. V-I characteristics of LED.

7. Study of B-H curve of a ferro magnetic material.
8. Study of P-E loop of given ferroelectric crystal.
9. Determination of dielectric constant of a given material.
10. Determination of frequency of AC mains using sonometer.

(Note: Any 8 experiments are to be performed.)

TEXT BOOKS:

1. Engineering Physics Laboratory Manual/Observation, Physics Faculty of VNRVJIET
2. A Textbook of Practical Physics, S. Balasubramanian, M. N. Srinivasan, S. Chand Publishers, 2017

REFERENCES:

1. Engineering Physics Lab Manual, Dr. Y. Aparna & Dr. K. Venkateswarao, V. G. S. Book links
2. Engineering Physics Practicals, Dr. B. Srinivasa Rao, V. K. V. Krishna, K. S. Rudramamba, University, Science Press, 2nd Edition, Imprint of Laxmi Publications

ONLINE RESOURCES:

1. https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=343&cnt=1
2. https://vlab.amrita.edu/index.php?sub=1&brch=280&sim=1518&cnt=1

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES2CS101) PROGRAMMING FOR PROBLEM SOLVING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To gain a working knowledge of C programming to write modular, efficient and readable C programs by Identifying the structural elements and layout of C source code
- To declare and manipulate single and multi-dimensional arrays of the C data types and derived data types like structures, unions
- To use functions from the portable C library and to describe the techniques for creating program modules using functions and recursive functions · To manipulate character strings in C programs. Utilize pointers to efficiently solve problems

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Design, Develop and debug programs by applying suitable C language constructs
CO-2: Implement solutions using arrays, strings, pointers, and user-defined data types
CO-3: Apply modular programming principles and handle data through file operations
CO-4: Perform low level operations for interfacing with memory
CO-5: Translate problem statements into executable code through hands-on programming practice

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	1	2	-	-	1	2	2	-	3	2	3
CO-2	3	3	3	1	2	-	-	1	2	2	-	3	2	2
CO-3	3	3	3	2	2	1	1	2	2	2	1	2	3	2
CO-4	3	3	3	2	2	1	1	2	2	2	1	3	3	3
CO-5	3	3	3	2	2	1	1	2	2	2	2	2	3	3

LIST OF EXPERIMENTS:

WEEK 1:

a) Programs on input, output statements
b) Programs on various operators
c) Programs on expression evaluation

WEEK 2:

a) Program that converts one given data type to another using auto conversion and casting. Take the values from standard input.
b) Programs on conditional statements- Simple if, if-else, Nested if-else, Else-if ladder, switch case

WEEK 3:

a) Programs on simple loops- while, for, do. while
b) Programs on Nested loops- while, for, do. while
c) Programs to understand goto, break, continue

WEEK 4:

a) Programs on 1-D arrays
b) Programs on linear, binary searching
c) Programs on bubble, selection and insertion sorting

WEEK 5:

a) Programs on 1-D strings
b) Programs using string handling functions

WEEK 6:

a) Programs on 2-D arrays
b) Programs on 2-D strings

WEEK 7:

a) Programs on user defined functions
b) Programs on passing arrays and strings to functions

WEEK 8: Internal Lab Exam -1

WEEK 9:

a) Programs on recursion
b) Programs on structures – simple structure, array of structures, array within structure, nested structure
c) Programs on Unions

WEEK 10:

a) Programs on pointers to variables
b) Programs on pointers to arrays(1-D, 2-D)

WEEK 11:

a) Program to understand call by value and call by address
b) Programs on pointers to strings
c) Programs on pointers to structure
d) Programs using malloc, calloc , realloc, free

WEEK 12:

a) Programs on macros, file inclusion, enum , typedef
b) Programs on sequential file accessing

WEEK 13:

a) Programs on error handling functions in files
b) Programs on Random file accessing
c) Programs on command line arguments

WEEK 14:

Lab Internal Exam -2

TEXT BOOKS:

1. The C Programming Language, Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall of India
2. C Programming and Data Structures, B. A. Forouzan and R. F. Gilberg, 3rd Edition, Cengage Learning
3. C: The Complete Reference, Herbert Schildt, 4th Edition, McGraw-Hill

REFERENCES:

1. Problem Solving and Program Design in C, Jeri R. Hanly and Elliot B. Koffman, 7th Edition, Pearson
2. Computer Fundamentals and C, E. Balagurusamy, 2nd Edition, McGraw-Hill
3. Let us C, Yashwant Kanetkar, 18th Edition, BPB
4. How to Solve it by Computer, R. G. Dromey, Pearson, 16th Impression 5. Programming in C, Stephen G. Kochan, 4th Edition, Pearson Education

ONLINE RESOURCES:

1. https://nptel.ac.in/courses/106105171
2. https://ugcmoocs.inflibnet.ac.in/index.php/courses/view_ug/307

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25HS2EN101) ENGLISH LANGUAGE AND COMMUNICATION SKILLS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

INTRODUCTION:
The aim of the English Language and Communication Skills Lab (ELCS Lab) is to develop the communicative skills of learners to function productively in academic and professional settings. This course focuses on developing Listening Skills, Reading Skills, Speaking Skills, Information transfer, Vocabulary, and pronunciation.

Professional students need reading strategies to explore scientific information and process the information. Further, they need to present and share this information in speaking and/or writing with their peers and scientific community. In addition to these, they also need good vocabulary for the presentation of their ideas. Practice is given not only in processing information to identify main ideas and relevant facts, to compare and contrast information and finally draw conclusions but also in presenting the same in speaking and writing.

COURSE OBJECTIVES:
- To train students to use neutral accent through phonetic sounds, symbols, stress and intonation
- To provide practice in vocabulary usage & grammatical construction
- To provide ample practice in LSRW skills and train the students in oral presentations, public speaking, role play, and situational dialogue
- To provide practice in defining technical terms and describing processes
- To equip students with excellent writing skills and information transfer skills

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Speak fluently with a neutral accent
CO-2: Use contextually apt vocabulary and sentence structures
CO-3: Make Presentations with great confidence
CO-4: Define technical terms and describe processes
CO-5: Write accurately, coherently, and lucidly

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	1	2	2	2	-	-	-	2	3	-	2	-	2	-
CO-2	1	2	2	2	3	3	2	2	3	2	2	1	3	1
CO-3	-	1	2	1	-	1	2	2	3	2	2	-	2	1
CO-4	2	2	2	2	2	-	-	1	3	2	2	1	3	-
CO-5	1	2	1	2	2	2	2	1	3	3	3	-	2	1

LIST OF EXPERIMENTS:

UNIT-I:

CALL

1. Common errors with Speech Sounds – Identify elements of Mother Tongue Influence,
2. Listening Skills – Types of Listening- Barriers – Active Listening Practice
3. Reading Skills -Skimming & Scanning-Reading for Gist -Reading for Specific Details

Oral Communication Lab:

1. Non-Verbal Communication
2. Formal and Informal English, Small Talk
3. Greetings – Introducing oneself & others - Elevator Pitch

UNIT-II:

CALL

1. **Activity:** Listening for information
2. Listening comprehension
3. **Reading Skills:** Reading comprehension-contextual vocabulary

Oral Communication Lab

1. Participatory Conversation
2. Role Playing in Formal Situations, Situational Dialogues – Two-Way (E. g. Elevator Pitch)
3. Media Etiquette, Conference Etiquette

UNIT-III:

CALL

1. **Phonetics:** Errors in Pronunciation, Neutralizing Mother Tongue Influence
2. **Listening Skills:** Listening for Gist
3. **Reading Skills:** SQ3R Method, Inferring

Oral Communication Lab

1. Describing Objects / Situations/ Places, People, Events,
2. Process Description

UNIT-IV:

CALL

1. **Listening Skills:** Techniques for Listening –Listening for Specific Details
2. **Reading Skills:** Making inferences

Oral Communication Lab

1. Story Telling; Story STAR, Sequencing, Creativity
2. Telling and Retelling Stories- Collage

UNIT-V:

CALL

1. **Listening Skills:** Listening for Evaluation, Writing the Summary
2. **Prompt Engineering:** Generating Appropriate Prompts, Keyword Extraction
3. Definition of a Technical Term; System Flow

Oral Communication Lab

1. Report Presentation
2. Short Presentation

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES2IT101) IT WORKSHOP

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Basic Computer Knowledge, Familiarity with Operating Systems, Typing & File Handling Skills, Basic Internet Skills, Fundamental Knowledge of Office Tools, Logical Thinking & Problem-Solving Ability, Willingness to Learn New Tools, and Openness to explore both proprietary (MS Office) and open-source (Linux, LaTeX) platforms

COURSE OBJECTIVES:

- To provide fundamental knowledge of computer hardware components and develop hands-on skills in assembling, disassembling, and configuring personal computers
- To train students in installing and managing multiple operating systems (Windows and Linux) and configuring systems for dual boot
- To impart practical knowledge of networking concepts, including LAN connectivity, TCP/IP configuration, and safe internet practices (cyber hygiene)
- To develop skills in creating professional documents using LaTeX and MS Word with advanced formatting, tables, images, and editing tools
- To enhance analytical, visualization, and presentation skills using MS Excel for data analysis and MS PowerPoint for effective communication

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explain the components of a computer system, identify CPU peripherals, and describe their functions through block diagrams

CO-2: Assemble and disassemble a personal computer and configure operating systems (Windows/Linux) including dual boot installation

CO-3: Configure TCP/IP settings, troubleshoot internet connectivity, and apply browser customizations with appropriate cyber hygiene measures

CO-4: Assess and compare document creation using LaTeX and MS Word by applying formatting, styles, tables, and mail merge features to meet professional documentation standards

CO-5: Design and develop analytical reports using MS Excel (formulas, charts, conditional formatting) and professional presentations in PowerPoint with integrated multimedia and master layouts

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	2	1	1	-	1	-	2	3	2	1
CO-2	3	2	3	2	3	2	1	2	1	1	2	3	3	2
CO-3	2	3	2	2	3	3	2	2	1	1	2	3	3	2
CO-4	2	2	2	1	2	1	1	1	3	2	2	2	2	3
CO-5	2	2	3	2	3	1	1	2	3	2	2	3	3	3

LIST OF EXPERIMENTS:

WEEK 1 to 4: PC Hardware

Task 1: Identify the peripherals of a computer, components in a CPU and its functions. Draw the block diagram of the CPU along with the configuration of each peripheral and submit it to your instructor.

Task 2: Every student should disassemble and assemble the PC back to working condition. Lab instructors should verify the work and follow it up with a Viva. Also, students need to go through the video which shows the process of assembling a PC. A video would be given as part of the course content.

Task 3: Every student should individually install MS windows Operating system into Virtual machine. Lab instructor should verify the installation and follow it up with a Viva.

Task 4: Every student should install Linux into Virtual Machine. Lab instructors should verify the installation and follow it up with a Viva

WEEK 5 & 6: Internet & World Wide Web

Task1: Orientation & Connectivity Boot Camp: Students should get connected to their Local Area Network and access the Internet. In the process they configure the TCP/IP setting. Finally, students should demonstrate, to the instructor, how to access the websites and email. If there is no internet connectivity preparations need to be made by the instructors to simulate the WWW on the LAN.

Task 2: Web Browsers, Surfing the Web: Students customize their web browsers with the LAN proxy settings, bookmarks, search toolbars and pop-up blockers.

Task 3: Search Engines & Netiquette: Students should know what search engines are and how to use the search engines. A few topics would be given to the students for which they need to search on Google. This should be demonstrated to the instructors by the student.

Task 4: Cyber Hygiene: Students would be exposed to the various threats on the internet and would be asked to configure their computer to be safe on the internet. They need to customize their browsers to block pop ups, block active x downloads to avoid viruses and/or worms.

WEEK 7 & 8: LaTeX and WORD

Task 1: Word Orientation: overview and Importance of LaTeX and Microsoft (MS) office, Using LaTeX and word – Accessing, overview of toolbars, saving files, Using help and resources, rulers, format painter in word.

Task 2: Using LaTeX and Word to create a project certificate. Features to be covered: - Formatting Fonts in word, Drop Cap in word, Applying Text effects, Using Character Spacing, Borders and Colors, Inserting Header and Footer, Using Date and Time option in both LaTeX and Word.

Task 3: Creating project abstract Features to be covered: -Formatting Styles, inserting table, Bullets and Numbering, Changing Text Direction, Cell alignment, Footnote, Hyperlink, Symbols, Spell Check, Track Changes.

Task 4: Creating a Newsletter: Features to be covered: - Table of Content, Newspaper columns, Images from files and clipart, Drawing toolbar and Word Art, Formatting Images, Textboxes, Paragraphs and Mail Merge in word.

WEEK 9 & 10: Excel Orientation

Using Excel – Accessing, overview of toolbars, saving excel files, Using help and resources.

Task 1: Creating a Scheduler - Features to be covered: Gridlines, Format Cells, Summation, auto fill, Formatting Text

Task 2: Calculating GPA - Features to be covered: - Cell Referencing, Formulae in excel – average, std. deviation, Charts, Renaming and Inserting worksheets, Hyper linking, Count function, LOOKUP/VLOOKUP
Task 3: Split cells, freeze panes, group and outline, Sorting, Boolean and logical operators, Conditional formatting

WEEK 11 & 12: PowerPoint
Task 1: Students will be working on basic power point utilities and tools which help them create basic PowerPoint presentations. PPT Orientation, Slide Layouts, Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Lines and Arrows in PowerPoint.
Task 2: Interactive presentations - Hyperlinks, Inserting –Images, Clip Art, Audio, Video, Objects, Tables and Charts.
Task 3: Master Layouts (slide, template, and notes), Types of views (basic, presentation, slide slotter, notes etc.), and Inserting – Background, textures, Design Templates, Hidden slides.

WEEK 13: Case Study:
Design and manage Conference Proceedings using MS Word, Excel, and PowerPoint.
MS Word: For drafting, formatting, and compiling research papers, abstracts, and official documents into a structured proceedings booklet.
MS Excel: For maintaining participant details, registrations, schedules, and generating analytical reports such as attendance and fee collection.
MS PowerPoint: For preparing and delivering keynote and technical session presentations, ensuring effective visual communication during the conference.

WEEK 14: Lab Internal Exam

TEXT BOOKS:

1. Comdex Information Technology Course Tool Kit, Vikas Gupta, Wiley Dreamtech
2. The Complete Computer Upgrade and Repair Book, 3rd Edition Cheryl A. Schmidt, Wiley Dreamtech
3. PC Hardware - A Handbook, Kate J. Chase, PHI

REFERENCES:

1. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education
2. LaTeX Companion, Leslie Lamport, PHI/Pearson
3. IT Essentials PC Hardware and Software Companion Guide, David Anfinson and Ken Quamme, 3rd Edition, CISCO Press, Pearson Education
4. IT Essentials PC Hardware and Software Labs and Study Guide, Patrick Regan, 3rd Edition, CISCO Press, Pearson Education

ONLINE RESOURCES:
1. https://www.youtube.com/watch?v=lhX0fOUYd8Q
2. https://www.overleaf.com/learn/latex/Tutorials
3. https://support.mozilla.org/en-US/products/firefox/privacy-and-security
4. https://www.cisa.gov/stay-safe-online

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS1MT106) ORDINARY DIFFERENTIAL EQUATIONS, LAPLACE TRANSFORM AND VECTOR CALCULUS

TEACHING SCHEME		
L	T/P	C
2	1	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematical Knowledge at Pre-University level

COURSE OBJECTIVES:
- To learn methods of solving the differential equations of first and higher order
- To learn moncept, properties of Laplace transforms
- To learn solving ordinary differential equations using Laplace transforms techniques
- To learn the physical quantities involved in engineering field related to vector valued functions
- To know the basic properties of vector valued functions and their applications to line, surface and volume integrals

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Identify whether the given differential equation of first order is exact or not
CO-2: Solve higher differential equation and apply the concept of differential equation to real world problems
CO-3: Use the Laplace transforms techniques for solving ODE's
CO-4: Evaluate the line, surface and volume integrals and converting them from one to another

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	2	1	-	-	-	-	-	2	1	1	-
CO-2	3	3	1	2	1	-	-	-	-	-	2	1	1	-
CO-3	3	2	1	2	2	-	-	-	-	-	2	2	1	-
CO-4	3	2	2	2	1	-	-	-	-	-	1	2	1	-

UNIT-I:
First Order ODE: Exact differential equations – Equations reducible to exact differential equations – linear and Bernoulli's equations – Orthogonal Trajectories (only in Cartesian Coordinates). Applications: Newton's law of cooling – Law of natural growth and decay.

UNIT-II:
Ordinary Differential Equations of Higher Order: Second order linear differential equations with constant coefficients: Non-Homogeneous terms of the type e^{ax}, sin ax, cos ax, polynomials in x, e^{ax} V(x) and x V(x) – Method of variation of parameters.

UNIT-III:

Laplace Transforms: Laplace Transforms: Laplace Transform of standard functions – First shifting theorem – Laplace transforms of functions multiplied by 't' and divided by 't' – Laplace transforms of derivatives and integrals of function – Evaluation of integrals by Laplace transforms – Laplace transform of periodic functions – Inverse Laplace transform, convolution theorem (without proof). Applications: solving Initial value problems (Linear ODEs with constant coefficients) by Laplace Transform.

UNIT-IV:

Vector Differentiation: Vector point functions and scalar point functions – Gradient – Directional derivatives -Divergence and Curl - Solenoidal and Irrotational vectors - Scalar potential functions – Vector Identities (without proof).

UNIT-V:

Vector Integration: Line, Surface and Volume Integrals. Theorems of Green, Gauss and Stokes (without proofs) and their applications.

TEXT BOOKS:

1. Higher Engineering Mathematics, B. S. Grewal, 36th Edition, Khanna Publishers, 2010
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, 5th Editon, Narosa Publications, 2016
3. Higher Engineering mathematics, B. V. Ramana, 11th Reprint, Tata McGraw-Hill, 2010

REFERENCES:

1. Advanced Engineering Mathematics, Erwin Kreyszig, 9th Edition, John Wiley & Sons, 2006
2. Advanced Engineering Mathematics, Dennis G. Zill, 6th Edition
3. A Textbook of Engineering Mathematics, N. P. Bali and Manish Goyal, Laxmi Publications, Reprint, 2008
4. Higher Engineering Mathematics, H. K. Dass and Er. Rajnish Verma, S. Chand

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS1CH101) CHEMISTRY FOR ENGINEERS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: General Chemistry and Basic Mathematics

COURSE OBJECTIVES:
- To analyze the quality of water for sustainable living
- To imbibe the conceptual knowledge of electrochemical processes and corrosion science
- To outline the importance of non-conventional energy sources and portable electronic devices
- To acquire knowledge about polymer science and its applications in various fields
- To recognize the significance of analytical techniques and advanced functional materials in engineering, industrial, environmental, and biomedical applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Assess the specification of water regarding its usage in domestic & Industrial scenarios
CO-2: Interpret electrode potentials and predict the suitable corrosion control methods in safeguarding the structures
CO-3: Recognize the transformations in energy sources & battery technology
CO-4: Analyze the efficacy of polymers in diverse applications
CO-5: Identify the use of spectroscopic techniques and advanced functional materials in varied applications

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	2	1	1	1	1	1	2	-	-	1
CO-2	3	2	2	1	1	1	1	1	1	1	2	-	-	1
CO-3	3	2	2	1	1	1	1	1	1	1	2	-	-	1
CO-4	3	2	2	1	1	2	1	1	1	1	2	-	-	1
CO-5	3	2	1	1	1	1	1	1	1	1	2	-	-	1

UNIT-I:
Water and its Treatment: Introduction- Hardness, types, degree of hardness and units – numerical Problems. Estimation of hardness of water by complexometric method. Potable water and its specifications (WHO) – Steps involved in the treatment of potable water – screening, sedimentation, coagulation, filtration and disinfection by chlorination and break-point chlorination.
Boiler Troubles: Scales, Sludges, Internal treatment of boiler feed water – Calgon conditioning, Phosphate conditioning, Colloidal conditioning. External treatment methods – Softening of water by Ion-exchange process. Membrane Separation –

Types of membranes [Microfiltration (MF), Ultrafiltration (UF), Nanofiltration (NF)], advantages and applications. Desalination of brackish water – Reverse osmosis.

UNIT-II:

Electrochemistry and Corrosion: Introduction – Electrochemical cell & galvanic cell, electrode potential, standard electrode potential, Nernst equation (no derivation). Types of electrodes – Reference electrodes – Construction, working principle, advantages, limitations and applications of Primary reference electrode (Standard Hydrogen Electrode – SHE), Secondary reference electrode (Calomel electrode) and Ion-selective electrode (Combined glass electrode).

Corrosion: Introduction – Definition, causes and effects of corrosion, Theories of corrosion – chemical and electrochemical theories of corrosion. Types of corrosion – galvanic, waterline and pitting corrosion. Factors affecting rate of corrosion – nature of metal (position, passivity, purity, areas of anode and cathode) & nature of environment (temperature, pH, humidity). Corrosion control methods – Cathodic protection Method – Sacrificial anode and impressed current cathodic protection methods.

UNIT-III:

Energy Sources and Battery Technology:

Fuels: Introduction and characteristics of good fuel, classification based on physical state.

Fossil Fuels: Petroleum – Refining of Crude oil, Cracking – Types of cracking – Moving bed catalytic cracking. Synthetic Fuels: Fischer-Tropsch process.

Gaseous fuels: Composition and uses of LPG, CNG and Hythane. Green Hydrogen (generation by water electrolysis – PEM, Alkaline, Solid Oxide).

Batteries: Introduction – Classification of batteries – Primary, secondary, reserve batteries and fuel cells (differences and examples). Construction, working and applications of Zn-air (primary battery) and Lithium-ion battery (secondary battery), Characteristic features and Advantages of Metal ion batteries (Sodium Ion Battery). Fuel Cells – Construction and applications of Proton Exchange Membrane Fuel Cell (PEMFC).

UNIT-IV:

Polymers: Definition – Classification of polymers – Types of polymerizations – Addition and condensation polymerization, differences between thermoplastics and thermosetting plastics. Plastics, Elastomers and Fibers: Synthesis, properties and applications of Teflon, Buna-S and Kevlar.

Fiber Reinforced Plastics (FRPs): Classification, features and applications.

Conducting Polymers: Definition, classification and applications.

Shape Memory Polymers: Definition, classification and applications.

Biodegradable polymers: Introduction – Definition Example: Features and applications of Polylactic acid (PLA).

UNIT-V:

Spectroscopic Applications and Advanced Functional Materials: Interpretative spectroscopic applications of UV-Visible spectroscopy for analysis of pollutants in dye industry, IR spectroscopy in night vision – security, Pollution Under Control – CO sensor (Passive Infrared detection), Raman spectroscopy (application) – Tumour detection in medical applications.

Advanced Functional Materials: Smart materials - Shape Memory Alloys (Example: Nitinol), Nanomaterials – Characterization by STM & AFM.

Self-healing Materials: Definition, working principle and applications.

Characteristic Features of Biosensing Materials: Example: Amperometric Glucose monitoring sensor.

TEXT BOOKS:

1. Engineering Chemistry, P. C. Jain and M. Jain, Dhanpat Rai and Company, 2010
2. Engineering Chemistry, Rama Devi, Dr. P. Aparna and Rath, Cengage Learning, 2025
3. Engineering Chemistry, Thirumala Chary, Laxminarayana & Shashikala, Pearson Publications, 2020

REFERENCES:

1. Engineering Chemistry, Shashi Chawla, Dhanpat Rai and Company, 2011
2. Engineering Chemistry, Shikha Agarwal, Cambridge University Press, 2015
3. Textbook of Engineering Chemistry, Jaya Shree Anireddy, Wiley Publications
4. Engineering Analysis of Smart Material Systems, Donald J. Leo, Wiley, 2007
5. Challenges and Opportunities in Green Hydrogen, Paramvir Singh, Avinash Kumar Agarwal, Anupma Thakur, R. K. Sinha, Springer Nature, 2024

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES1EE102) BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematics, Physics

COURSE OBJECTIVES:
- To understand the use of electrical energy in different engineering fields
- To analyze electrical circuits using different network reduction techniques
- To know the working, construction of electrical machines and installations
- To learn principles of operation, construction and characteristics of various electronic devices

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Appreciate the role of electrical energy in various engineering branches and analyze the DC circuits using various network reduction techniques
CO-2: Analyze the various electrical parameters of AC circuits with R-L-C elements
CO-3: Understand the operation of various Electrical Machines & Installations
CO-4: Apply basic electronics device in real time applications

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	1	1	-	-	-	-	1	1	2	-
CO-2	2	3	2	1	1	1	-	-	-	-	1	1	2	-
CO-3	2	3	2	1	1	1	-	-	-	-	1	1	2	-
CO-4	2	3	2	1	1	1	-	-	-	-	1	1	2	-

UNIT-I:
Introduction to Electrical Energy and DC Circuits: The role of Electrical Energy in modern life and various engineering branches- Circuit Concept – Types of Elements R-L-C parameters – Voltage and Current sources – Independent and dependent sources- Kirchhoff's laws – network reduction techniques: series, parallel, series parallel, star/delta transformations, Superposition theorem, Thevenin's theorem

UNIT-II:
Steady State AC Circuits: Representation of sinusoidal waveforms, average and RMS values, form factor and peak factor, phasor representation, Analysis of single-phase AC circuits consisting of R, L, C, series RL, RC, RLC combinations, real power, reactive power, apparent power, power factor - Three-phase balanced circuits, voltage and current relations in star and delta connections (Derivation only).

UNIT-III:

Electrical Machines: Transformer principle, types, Equivalent circuit, Regulation and Efficiency, DC generator principle, Emf equation, DC motor principle, Back emf, Three phase induction motor, types, principle, torque Slip characteristics, working principle of Synchronous generator, principle of operation of synchronous motor, stepper motor principle and applications.

UNIT-IV:

Diode and BJT: PN Junction Diode: Operation and Symbol, V-I Characteristics, Half Wave Rectifier, Full-Wave Rectifier (Centre Tap and Bridge Rectifiers), Zener Diode, V-I Characteristics

Bipolar Junction Transistor (BJT): Construction (NPN and PNP transistors), Transistor configurations (CC, CB and CE), Transistor as an Amplifier

UNIT-V:

Op-Amp and Potentiometers:

Operational Amplifier (IC 741): Basic information of Op-Amp, Pin Configuration of IC 741, Characteristics of Ideal Op-Amp, Block Diagram of Op-Amp, Applications of Op-Amp (only over view): Inverting and Non- Inverting Amplifier.

Potentiometer (Variable Resistor-Pot): Introduction to potentiometer, definition and working principle, symbol and circuit representation, Types of potentiometers (only over view)-Rotary, Linear and Digital.

TEXT BOOKS:

1. Basics of Electrical Energy for Engineers, V. Ramesh Babu, BS Publications, 2023
2. Basic Electrical Engineering, P. Ramana, M. Suryakalavathi, G. T. Chandra Sekhar, 1st Edition, S. Chand, 2018
3. Op-Amps Linear Integrated Circuits, 4th Edition, Ramakant A. Gayakwad, Prentice Hall of India, 2015

REFERENCES:

1. Engineering Circuit Analysis, William Hayt and Jack E. Kemmerly, 8th Edition, McGraw-Hill, 2013
2. Electrical and Electronics Technology, E. Hughes, 10th Edition, Pearson, 2010
3. Electrical and Electronics Measurements and Instrumentation, A. K. Sawhney, 3rd Edition, Dhanpat Rai & Co., 1983
4. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, 3rd Edition, Tata McGraw-Hill, 2010
5. The potentiometer Hand Book, User Guide to Cost Effective Applications

ONLINE RESOURCE:

1. https://nptel.ac.in/courses/117106108

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES1IT102) DATA STRUCTURES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Programming for Problem Solving

COURSE OBJECTIVES:
- To introduce various sorting techniques
- To demonstrate operations of linear and non-linear data structure
- To develop an application using suitable data structure

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand basic concepts of data structures and analyze computation complexity
CO-2: Solve the given problem using linear data structures
CO-3: Apply linear data structures to implement various sorting techniques
CO-4: Execute the given problem using non-linear data structures
CO-5: Implement dictionaries using hashing with basic operations and collision handling

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	-	3	-	-	-	2	2	-	2	2	2
CO-2	3	3	2	2	3	-	-	-	2	2	-	2	3	2
CO-3	3	3	2	2	3	-	-	-	2	2	-	2	3	2
CO-4	2	2	1	2	-	-	-	-	2	2	-	3	3	3
CO-5	3	3	3	2	3	2	2	2	3	3	1	3	3	3

UNIT-I:
Introduction to Data Structures: Abstract Data Types (ADT), Asymptotic Notations.
Linear Data Structures: Limitations of Array, Stacks -Stack ADT and its operations
Applications of Stacks: Recursion, Infix to postfix conversion, Postfix evaluation.

UNIT-II:
Linear Data Structures: Queues -Queue ADT, Types of Queues: Linear Queue, Circular Queue, Double ended queue, operations on each types of Queues
Linked Lists: Singly linked lists, Representation in memory, Operations: Traversing, Searching, insertion, Deletion from linked list, Linked representation of Stack and Queue.

UNIT-III:
Doubly Linked List, Circular Linked Lists: Operations-Traversing, Searching, insertion, Deletion.

Sorting: Objective and properties of different sorting algorithms: Quick Sort, Merge Sort, Radix Sort, Heap Sort.

UNIT-IV:

Trees: Basic Tree Terminologies, Different types of Trees: Binary Tree, Binary Search Tree, AVL Tree; Tree Operations on each of the trees and their algorithms with time complexities.

Priority Queue: Definition, Operations and their time complexities.

B-Trees: Definition, Operations.

UNIT-V:

Dictionaries: Definition, ADT, Linear List representation, operations- insertion, deletion and searching, Hash Table representation, Hash Functions-Division Method, Multiplication Method, Mid-Square Method, Folding Method, Collision Resolution Techniques-Separate Chaining, open addressing-linear probing, quadratic probing, double hashing, Rehashing.

Graphs: Graph terminology –Representation of graphs –Graph Traversal: BFS (breadth first search) –DFS (depth first search) –Minimum Spanning Tree.

TEXT BOOKS:

1. Fundamental of Data Structure, Horowitz and Sahani, Galgotia Publication
2. Data Structures Using C and C++, Aaron M. Tenenbaum, Moshe J. Augenstein, Yedidyah L. Angsam, 2nd Edition, PHI, 2009
3. Data Structures and Algorithms, Alfred V. Aho, John E. Hopperoft, Jeffrey D. Ullman

REFERENCES:

1. Algorithms, Data Structures, and Problem Solving with C++, Mark Allen Weiss, Addison-Wesley
2. How to Solve it by Computer, R. G. Dromey, 2nd Impression, Pearson Education
3. Introduction to Algorithms, Cormen, Leiserson and Rivest
4. Data Structures: A Pseudo-code Approach with C, Gilberg & Forouzan, Thomson Learning
5. Data Structures using C & C++, Ten Baum, Prentice-Hall International

ONLINE RESOURCES:

1. https://nptel.ac.in/courses/106102064
2. https://onlinecourses.swayam2.ac.in/cec19_cs04/preview

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES3ME101) ENGINEERING DRAWING

TEACHING SCHEME		
L	T/P	C
0	6	3

EVALUATION SCHEME				
D-D	SE	CP	SEE	TOTAL
10	20	10	60	100

COURSE OBJECTIVES:

- To understand the importance of engineering curves
- To learn to use the orthographic projections for points, lines and planes
- To understand the projections of solids in different positions
- To learn the importance of isometric projections and its conversions

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply the concepts of engineering curves and its construction using AutoCAD
CO-2: Solve the problems of projections of points, lines, planes in different positions using AutoCAD
CO-3: Solve the problems of projections of solids using AutoCAD
CO-4: Solve the problems on Conversion of Isometric views to Orthographic Views & Orthographic to Isometric Views using AutoCAD

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	-	-	3	2	-	-	-	-	2	2	1	1
CO-2	3	2	-	-	3	2	-	-	-	-	2	2	1	1
CO-3	3	2	-	-	3	2	-	-	-	-	2	2	1	2
CO-4	3	2	-	-	3	2	-	-	-	-	2	2	1	2

Introduction to AutoCAD Software:

The Menu System, Toolbars (Standard, Object Properties, Draw, Modify and Dimension), Drawing Area (Background, Crosshairs, Coordinate System), Dialog boxes and windows, Shortcut menus (Button Bars), The Command Line, The Status Bar, Different methods of zoom as used in CAD, Select and erase objects.

UNIT-I:

Introduction to Engineering Drawing: Principles of Engineering drawing and their significance and Conventions
Engineering Curves: Construction of Ellipse, Parabola and Hyperbola – General and Special methods; Cycloidal curves- Epicycloids and Hypocycloids

UNIT-II:

Orthographic Projections, Projections of Points & Straight Lines: Principles of Orthographic Projections – Conventions; Projections of Points in all positions; Projections of lines inclined to both the planes

UNIT-III:
Projections of Planes: Projections of Planes- Surface Inclined to both the Planes

UNIT-IV:
Projections of Regular Solids: Projections of Regular Solids inclined to both the Planes – Prisms, Pyramids, Cylinder and Cone

UNIT-V:
Isometric Projections: Principles of Isometric projection – Isometric Scale, Isometric Views, Conventions; Isometric Views of lines, Planes, Simple and Compound Solids Conversion of Isometric Views to Orthographic Views and Vice-versa, Conventions

TEXT BOOKS:

1. Engineering Drawing, N. D. Bhatt, 53rd Edition, Charotar Publishing House, 2016
2. Textbook on Engineering Drawing, K. L. Narayana & P. Kannaiah, Scitech Publishers, 2010
3. Engineering Drawing and Computer Graphics, M. B. Shah & B. C. Rana, Pearson Education, 2010

REFERENCE:

1. Mastering AutoCAD 2021 and AutoCAD LT 2021, George Omura and Brian C Benton (Auto CAD 2021), 1st Edition, John Wiley & Sons

ONLINE RESOURCES:
1. https://www.classcentral.com/course/swayam-engineering-graphics-5305
2. https://www.mooc-list.com/tags/engineering-drawing

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS2CH101) ENGINEERING CHEMISTRY LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Basic Knowledge of Volumetric Analysis and Mathematics

COURSE OBJECTIVES:
- To understand the preparation of standard solutions and handling of instruments
- To determine and evaluate the water quality
- To measure physical properties like absorption of light, surface tension, pH, conductance, and viscosity of various liquids
- To conduct and collect the experimental data using different laboratory techniques
- To summarize the data and find the applicability to real world scenario

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Learn and apply the basic laboratory methodologies for the preparation of the standard solutions and handling of instruments
CO-2: Estimate the ions / metal ions present in domestic and industrial water
CO-3: Utilize the instrumental techniques to assess the physical properties of oils and water
CO-4: Analyze the experimental data to predict solutions for complex engineering problems
CO-5: Apply the skills gained to address societal issues related to real world scenario

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	1	-	1	1	1	1	1	2	-	-	-
CO-2	3	2	2	2	1	2	1	1	1	1	2	-	-	-
CO-3	3	2	1	2	1	2	1	1	1	1	2	-	-	-
CO-4	3	2	2	2	1	2	1	1	1	1	2	-	-	1
CO-5	3	2	2	2	1	2	1	1	1	1	2	-	-	1

LIST OF EXPERIMENTS:

VOLUMETRIC ANALYSIS:
1. Estimation of hardness of water by complexometric method using EDTA.
2. Determination of chloride content in the given sample water using Argentometric method.
3. Determination of Iron using Potassium Dichromate.

COLORIMETRY:

4. Estimation of copper present in the given solution by colorimetric method.

CONDUCTOMETRY:

5. Conductometric titration of Acid vs Base.
6. Conductometric titration of mixture of strong acid and weak acid by strong base.

pH METRY:

7. Titration of Acid vs Base using pH metric method.
8. Determination of viscosity of sample oil by Redwood Viscometer-I.
9. Estimation of acid value of given lubricant oil.
10. Determination of surface tension of a liquid by drop method using Stalagmometer.

PREPARATIONS:

11. Preparation of Bakelite.
12. Preparation of Nylon-6,6.
13. Determination of rate of corrosion of mild steel in the presence and absence of inhibitor.
14. **VIRTUAL LAB EXPERIMENTS / CBP:**

- Construction of advanced Fuel cell and it's working.
- Smart materials for Biomedical applications
- Batteries for electrical vehicles.
- Functioning of solar cell (materials used) and its applications.

TEXT BOOKS:

1. Laboratory Manual on Engineering Chemistry, S. K. Bhasin and Sudha Rani, Dhanpat Rai Publications
2. College Practical Chemistry, V. K. Ahluwalia, Sunitha Dhingra, Adargh Gulati, University Press
3. Practical Chemistry, Dr. O. P. Pandey, D. N. Bajpai, and Dr. S. Giri, S. Chand

REFERENCES:

1. Vogel's Textbook of Quantitative Chemical Analysis, G. N. Jeffery, J. Bassett, J. Mendham and R. C. Denny, Longmann, ELBS
2. Advanced Practical Physical Chemistry, J. D. Yadav, Goel Publishing House
3. Practical Physical Chemistry, B. D. Khosla, R. Chand and Sons
4. Lab Manual for Engineering Chemistry, B. Ramadevi and P. Aparna, S. Chand, 2022
5. Vogel's Textbook of Practical Organic Chemistry, 5th Edition

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES2IT102) DATA STRUCTURES LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Programming for Problem Solving

COURSE OBJECTIVES:
- To impart the basic concepts of data structures
- To learn the concepts of sorting
- To understand the basic concepts about stacks, queues, lists
- To know the concepts of trees and graphs

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Implement all operations on different linear data structures
CO-2: Implement Applications of Linear data structures
CO-3: Apply various sorting techniques
CO-4: Develop all operations on different Non-linear data structures
CO-5: Use appropriate data structure for any given problem

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	-	2	-	-	-	1	1	-	2	2	1
CO-2	3	3	2	1	2	-	-	-	2	2	-	3	3	2
CO-3	3	3	2	1	2	-	-	-	2	2	-	3	3	2
CO-4	2	3	2	1	2	-	-	-	2	2	-	3	3	3
CO-5	3	3	3	2	3	2	1	2	3	3	1	3	3	3

LIST OF EXPERIMENTS:

WEEK 1:
Implement Stack using Array

WEEK 2:
a) Program to convert infix expression to postfix expression.
b) Program to postfix evaluation.

WEEK 3:
Implement the following
a) Linear Queue using Array   b) Circular Queue using Array

WEEK 4:
Implement Deque using Array

WEEK 5:
Implement Single Linked List operations

WEEK 6:
Implement following
a) Circular Linked List Operations b) Double Linked List Operations

WEEK 7:
Implement following
a) Stack using Linked List b) Queue using Linked List

WEEK 8:
Lab Internal – 1

WEEK 9:
a) Implement Priority Queue
b) Implement Double ended Queue using Double Linked List

WEEK 10:
Implement following sorting techniques
a) Merge b) Heap c) Quick d) Radix

WEEK 11:
Implement BST operations

WEEK 12:
Implement following Graph traversals
a) BFS b) DFS

WEEK 13:
Implement following Hashing Techniques
a) Separate Chaining b) Linear Probing c) Quadratic Probing

WEEK 14:
Internal Lab – 2

TEXT BOOKS:
1. Fundamental of Data Structure, Horowitz and Sahani, Galgotia Publication
2. Data Structures Using C and C++, Aaron M. Tenenbaum, Moshe J. Augenstein, Yedidyah L. Angsam, 2nd Edition, PHI, 2009
3. Data Structures and Algorithms, Alfred V. Aho, John E. Hopperoft, Jeffrey D. Ullman

REFERENCES:
1. Algorithms, Data Structures, and Problem Solving with C++, Mark Allen Weiss, Addison-Wesley Publishing Company
2. How to Solve it by Computer, 2nd Impression, R. G. Dromey, Pearson Education
3. Introduction to Algorithms, Cormen, Leiserson and Rivest

4. Data Structures: A Pseudo-code Approach with C, Gilberg & Forouzan, Thomson Learning
5. Data Structures using C & C++, Ten Baum, Prentice-Hall International

ONLINE RESOURCES:

1. https://nptel.ac.in/courses/106102064
2. https://onlinecourses.swayam2.ac.in/cec19_cs04/preview

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES2CY101) PYTHON PROGRAMMING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To install and run the Python interpreter & IDE's
- To learn control structures
- To understand advanced data Types, Functions in Python
- To handle exceptions, files and Oops concepts in Python
- To understand Python Advanced modules like numpy

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply comprehensions, different Decision-Making statements and Functions
CO-2: Implement various data types like lists, tuples, strings
CO-3: Use different File handling operations and Maps
CO-4: Apply object oriented programming in Python types
CO-5: Use Numpy in developing mathematical applications

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	2	-	-	-	-	1	2	3	2	3
CO-2	3	2	3	2	2	-	-	-	-	2	2	2	1	2
CO-3	3	3	3	3	2	-	2	-	-	3	2	2	3	1
CO-4	3	3	2	3	3	-	-	-	-	3	3	2	2	2
CO-5	3	2	3	3	3	-	-	2	-	3	3	2	3	2

LIST OF EXPERIMENTS:

WEEK 01:

a) Install Python and configure Interpreter & IDE like Jupyter, google colab.
b) Write and execute your first Python program to display "Hello, Python!".
c) Demonstrate interactive mode and script mode.
d) Use command-line arguments to run Python scripts.
e) Write a program to find the largest element among three Numbers.
f) Write a program to print the sum of all the even numbers in the range 1 - 50 and print the even sum.
g) Write a Program to display all prime numbers within an interval of given X1 & X2.

WEEK 02:

a) Program to demonstrate Python's built-in data types.

b) Type conversion using int(), float(), str(), etc.
c) Calculate Simple Interest and Area of a Circle.
d) Program to swap two numbers without using a third variable.

WEEK 03:

a) Write a program to print the following patterns using loop:
   *
   **
   ***
   ****
b) Write a program to print multiplication table of a given number X1 to range X2.
c) Grade calculation using if-elif-else.
d) Determine the type of a triangle (equilateral, isosceles, scalene).

WEEK 04:

a) Write a program to find the length of the string without using any library functions.
b) Write a program to check if two strings are anagrams or not.
c) Write a program to check if the substring is present in a given string or not.
d) Write a program to perform the given operations on a list:
   i. add ii. insert iii. slicing
e) Write a program to perform any 5 built-in functions by taking any list.
f) Write a program to get a list of the even numbers from a given list of numbers. (use only comprehensions)

WEEK 05:

a) Write a program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples.
b) Write a program to return the top 'n' most frequently occurring chars and their respective position.
c) Write a program on counts. e. g. aaaaabbbbcccc, it should return ex:
   [(a 5) (b 4) (c 4)]
d) Write a program to count the number of vowels in a string.
e) Write a program that displays which letters are present in both strings.
f) Write a program to sort given list of strings in the order of their vowel counts.

WEEK 06:

a) Write a program to check if a given key exists in a dictionary or not.
b) Write a program to add a new key-value pair to an existing dictionary.
c) Write a program to sum all the items in a given dictionary.
d) Demonstrate functions with different argument types: positional, keyword, default, variable- length.
e) Program to show local vs global variable behavior.
f) Recursive function to calculate factorial.
g) Recursive Fibonacci sequence generator.

WEEK 07:

a) Write a program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper case words from source must be lowered.
b) Write a program to find the most frequent words in a text. (read from a text file).

WEEK 08:

Lab internal -1

WEEK 09:

a) Program using try, except, else, and finally.
b) Raise and handle custom exceptions.
c) Read and write data to a text file.
d) Count the number of words and lines in a text file.

WEEK 10:

a) Write a Python class named Person with attributes name, age, weight (kgs), height (ft) and takes them through the constructor and exposes a method get_bmi_result() which returns one of "underweight", "healthy", "obese".
b) Write a Python class named Circle constructed by a radius and two methods which will compute the area and the perimeter of a circle.

WEEK 11:

a) Accept two lists, one list represents temperatures in Fahrenheit and another list represents temperatures in Celsius. Perform map operations Fahrenheit-Celsius and Celsius- Fahrenheit using lambda
b) Create a Fibonacci sequence that contains 'N' terms, and filter only even terms using lambda
c) Write a program to find the number of rows in a text file using Generator and yield.
d) Find Sum of Squares of 1 to n numbers using Generator Expressions.

WEEK 12:

Let's take a password as a combination of alphanumeric characters along with special characters, and check whether the password is valid or not with the help of a few conditions.

Conditions for a valid password are:

- Should have at least one number.
- Should have at least one uppercase and one lowercase character.
- Should have at least one special symbol.
- Should be between 6 to 20 characters long.

Input : Gvpce12#   Output : Password is valid.
Input : asd123   Output : Invalid Password !!

WEEK 13:

a) Write a Python Program to demonstrate numpy arrays creation using array () function.
b) Write a python program to demonstrate use of ndim, shape, size, dtype.
c) Python program to demonstrate basic slicing, integer and Boolean indexing.
d) Write a python program to find min, max, sum, cumulative sum of array.
e) Create two single dimensional NumPy arrays, one is height, and another is weight, calculate BMI (weight/height**2) and keep all BMI values in another NumPy Array. Calculate mean, median, and standard deviation of BMI values.

WEEK 14 :

Lab internal 2

TEXT BOOKS:

1. Python with Machine Learning, A. Krishna Mohan, T. Murali Mohan & Karunakar, 1st Edition, S. Chand Publications, 2019
2. Introduction to Programming using Python, Y. Daniel Liang, 1st Edition, Pearson Publications, 2017

REFERENCES:

1. Python Programming A Modular Approach, Sheetal Taneja, 1st Edition, Pearson Publications, 2017
2. Effective Python: 59 Specific Ways to Write Better Python, I/C, Brett Slatkin (C), 1st Edition Pearson Publications, 2015
3. Programming and Problem Solving with Python, Ashok Namdev Kamathane and Amit Ashok Kamathane, 1st Edition, McGraw-Hill Education, 2017

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES2ME101) ENGINEERING WORKSHOP

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To know the different popular manufacturing process
- To gain a good basic working knowledge required for the production of various engineering products
- To provide hands on experience about use of different engineering materials, tools, equipment and processes those are common in the engineering field
- To identify and use marking out tools, hand tools, measuring equipment and to work to prescribed tolerances

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand various types of manufacturing processes
CO-2: Fabricate/make components from wood and steels through hands on experience
CO-3: Understand different machining processes like turning, drilling, tapping, etc.
CO-4: Understand electrical and electronic components and their assembly

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	-	-	1	-	-	-	-	3	3	2	-	2	1	-
CO-2	-	-	1	-	-	-	-	3	3	2	-	2	1	-
CO-3	-	-	1	-	-	-	-	3	3	2	-	2	1	-
CO-4	-	-	1	-	-	-	-	3	3	2	-	1	1	-

LIST OF EXPERIMENTS:

1. TRADES FOR EXERCISES:

i) **Carpentry:** Cross lap joint, Mortise & tenon joint
ii) **Fitting:** L-fitting, Square fitting
iii) **Tin smithy:** Rectangular tray, Rectangular scoop
iv) **Arc welding:** Lap joint, Butt joint
v) **House Wiring:** Single lamp connection, Staircase connection
vi) **Black smithy:** Round to Square, U-Hook
vii) **Machine shop:** Step turning on lathe, Drilling & tapping

2. TRADES FOR DEMONSTRATION:

Plumbing, 3D printing, Power tools in construction and wood working.

TEXT BOOKS:

1. Workshop Manual, P. Kannaiah and K. L. Narayana, 3rd Edition, Scitech, 2015
2. Elements of Workshop Technology Vol. 1 & 2, S. K. Hajra Choudhury, A. K. Hajra Choudhury and Nirjhar Roy, 13th Edition, Media Promoters & Publishers, 2010

REFERENCES:

1. Manufacturing Engineering and Technology, Serope Kalpakjian, Steven R. Schmid, 4th Edition, Pearson Education India Edition, 2002
2. Manufacturing Technology-I, S. Gowri, P. Hariharan and A. Suresh Babu, Pearson Education, 2008
3. Processes and Materials of Manufacture, Roy A. Lindberg, 4th Edition, Prentice Hall India, 1998
4. Manufacturing Technology Vol-1 & 2, P. N. Rao, Tata McGraw-Hill, 2017

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES2EE102) BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Mathematics, Physics and Basic Electrical Engineering

COURSE OBJECTIVES:
- To recognize different circuit reduction techniques
- To understand the construction of electrical equipment and operation
- To practice the techniques to control and assess electrical machines and installations
- To know operation and characteristics of various electronic devices

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply different network reduction techniques to solve and analyze electrical circuits
CO-2: Realize the compatibility of electrical machines in different engineering fields
CO-3: Control different electrical machines and evaluate their performance
CO-4: Identify different basic electronics device in real time applications

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	1	1	-	-	-	-	1	1	2	-
CO-2	2	3	2	1	1	1	-	-	-	-	1	1	2	-
CO-3	2	3	2	1	1	1	-	-	-	-	1	1	2	-
CO-4	2	3	2	1	1	1	-	-	-	-	1	1	2	-

LIST OF EXPERIMENTS:

1. Demonstration of safety precautions, measuring instruments, electrical and electronic components
2. Demonstration of LT switchgear components
3. Verification of KVL & KCL
4. Verification of the Superposition theorem
5. Verification of Thevenin's theorem
6. Analysis of series RL, RC, and RLC circuits
7. Load test on 1- φ transformer
8. Speed control of separately excited DC motor
9. PN Junction Diode Characteristics
10. Full Wave Rectifiers without filter
11. Transistor CE Characteristics (Input and Output)
12. Op-Amp as Inverting and Non-Inverting Amplifier

TEXT BOOKS:

1. Basic Electrical Engineering, D. C. Kulshreshtha, 2nd Edition, TMH, 2019
2. Basic Electrical Engineering, P. Ramana, M. Suryakalavathi, G. T. Chandra Sekhar, 1st Edition, S. Chand, 2018
3. Electronic Devices and Circuits, S. Salivahanan and N. Suresh Kumar, 3rd Edition TMH, 2019

REFERENCES:

1. Electrical and Electronics Technology, E. Hughes, 10th Edition, Pearson, 2010
2. Electrical Engineering Fundamentals, Vincent Deltoro, 2nd Edition, Prentice Hall India, 1989
3. Electrical and Electronics Measurements and Instrumentation, A. K. Sawhney, 3rd Edition, Dhanpat Rai & Co., 1983
4. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, 3rd Edition, Tata McGraw-Hill, 2010
5. Engineering Circuit Analysis, William Hayt and Jack E. Kemmerly, 8th Edition, McGraw-Hill, 2013

ONLINE RESOURCE:

1. https://nptel.ac.in/courses/117106108

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25MN6HS102) ENVIRONMENTAL SCIENCE

TEACHING SCHEME		
L	T/P	C
2	0	0

EVALUATION SCHEME		
SE-I	SE-II	TOTAL
50	50	100

COURSE PRE-REQUISITES: Basic knowledge on environmental issues

COURSE OBJECTIVES:
- To recognize the principles of environment
- To identify the sources and causes of Environmental Pollution
- To analyze the causes of climate change and identify climate change mitigation and adaptation technologies
- To list out the benefits in creating circular economy and sustainability
- To emphasize the Environmental conventions, policies and regulations

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Gain knowledge about principles of Environment and allied problems
CO-2: Identify the causes and suggest remedial measures for Environmental Pollution
CO-3: Appraise the mitigation measures and adaptation technologies related to climate change
CO-4: Familiarize with the importance of circular economy towards sustainability
CO-5: Relate and implement environmental policies and regulations

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	-	1	2	2	1	1	1	2	-	-	-
CO-2	2	2	1	-	1	2	2	1	1	1	2	-	-	-
CO-3	2	2	1	-	1	2	2	1	1	1	2	-	-	-
CO-4	2	2	1	-	1	2	2	1	1	1	2	-	-	-
CO-5	-	1	1	-	-	2	1	1	1	1	-	-	-	-

MODULE 1:
Introduction to Environmental Science: Importance and Scope of Environmental Science. Overview of Environment & its components. Ecosystem function and uses. Biodiversity – Types and Values.

MODULE 2:
Synergy With Environment: Relationship between environmental science & society – Influence of Industry & infrastructure on environment (Specific case studies). Sources, Impacts and prevention of air, water and soil pollution.

MODULE 3:
Climate Change: Science behind climate change – factors responsible for climate change, Role of greenhouse gases – Global temperature rise & its impact on environment & human health. Carbon footprint – Briefing on Paris agreement, Identify

key sectors for low carbon footprint. Climate change mitigation & adaptation strategies.

MODULE 4:

Moving Towards Sustainability: EIA and EMP - importance, Sustainable agriculture – Importance of Organic farming and hydroponics. Role of AI & IOT for efficient management of environmental issues – Health, air, water, and soil. Sustainable living practices – Principles and methods (Linear Economy vs circular economy). Green Buildings.

MODULE 5:

Policies and Regulations: International conventions: Montreal Protocol, COP15, Kyoto protocol (Salient Features). National Environmental Policies – Salient features of EPA act, Air act, Water Act, Forest and Wildlife act. National Green Tribunal Act (2010)- Features and Significance. Extended producer responsibility. Innovative approaches to waste management: Plastic, tyre and batteries.

TEXT BOOKS:

1. Living in the Environment, G. Tyler Miller/ Scott E. Spoolman, 20th Edition, Cengage Learning
2. Environmental Studies, Rajagopalan, Oxford University Press, 2005
3. Introduction to Climate Change, Andreas Schmittner, Oregon State University, 2018. https://open.oregonstate.education/climatechange/

REFERENCES:

1. Green Development: Environment and Sustainability in a Developing World, Bill Adams, 4th Edition, Routledge Publishers, 2021
2. Fixing Climate, Robert Kunzig & Wallace S. Broecker, Profile Books Publisher, 2009
3. Plastic Waste and Recycling-Environmental Impact, Societal Issues, Prevention and Solutions, 1st Edition, Edited by Trevor Letcher, Academic Press, 2020
4. Waste Management Practices: Municipal, Hazardous, and Industrial by John Pichtel, CRC Press, 2020
5. Environmental Science: A Global Concern, William P. Cunningham, Mary Ann Cunningham, McGraw Hill Education, 2022

REFERENCES:

1. https://www.google.co.in/books/edition/Waste_Management_Practices/pG3SBQAAQBAJ?hl=en&gbpv=1&pg=PA1&printsec=frontcover